

Conversation Redirection in Mental Health Therapy

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Abstract

Mental health therapy involves a complex conversation flow, in which patients and therapists continuously negotiate what should be talked about next. For example, therapists might shift the direction of the conversation to keep the therapeutic process on track and avoid stagnation, while patients might redirect the discussion towards issues that they want to focus on.

Our work introduces a probabilistic measure of the extent to which a certain utterance immediately *redirects* the flow of the conversation. We apply this new measure to characterize the development of patient-therapist relationships in a large online text-based therapy platform. Our analysis reveals (1) a temporal trend—with a relative increase of the patient control of the direction of the discussion as the relationship develops—and (2) a link between the ability of a patient to redirect the conversation and the likelihood that they will express dissatisfaction with the therapeutic relation and terminate it.

1 Introduction

Mental health therapy involves a complex flow of many conversational strategies, interventions, and topics (for a large range of operationalized categories see Imel et al. (2024)). For this reason, therapy is an ongoing negotiation between a patient and a therapist of what should be talked about, how it should be talked about, and by whom. Conversational bids by a party to shift the direction of the conversation (henceforth *redirections*) are a common and essential aspect of any therapeutic relationship.

Redirections have yet to be studied in a systematic and rigorous way due to their complex dialogic nature. As opposed to other commonly studied strategies and conversational acts (Malgaroli et al., 2023a) that tend to be executed by either the therapist (e.g., empathy (Sharma et al., 2020)) or the patient (e.g., change talk (Park et al., 2019)), redirection is a joint patient-therapist act: to be realized,

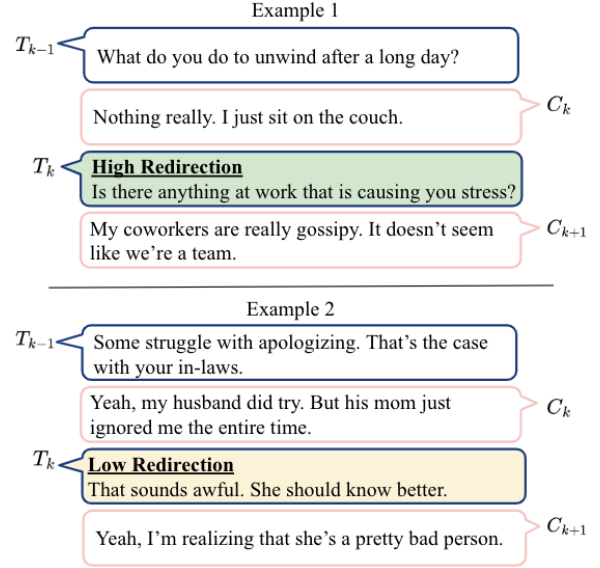


Figure 1: Utterances with high/low redirection effects. T_k refers to therapist's k -th utterance; C_k refers to patient's k -th utterance. (Rephrased to preserve privacy.)

a bid to redirect the conversation must be accepted by the other participant. In that sense, redirection is a truly dialogic conversational feature.

In this work, we introduce a measure to quantify the redirection effect of an utterance that explicitly takes into account this dialogic nature by jointly capturing the *intention* to change course of the conversation and the actual *realization* of this change through a reply that complies to this intention. Figure 1 illustrates an example of an utterance with high redirection effect, in which the therapist intends to switch the focus towards specific sources of stress, and the patient conforms by mentioning a source of frustration at work; had the patient resisted and said instead “I don't know, I'm just stressed all the time.” the redirection would not have been realized. The second example shows an utterance with low redirection effect due to the lack of intention to redirect. Note that in this work we aim to capture a localized phenomenon, cen-

tered around a specific utterance, as opposed to a conversation-level phenomenon (e.g., topic control (Nguyen et al., 2014, inter alia)).

This measure adds to the toolkit of computational methods available for studying therapeutic practice (Althoff et al., 2016; Imel et al., 2024; Margoli et al., 2023a). Its inherently dyadic nature is particularly well-suited for examining the therapeutic relationship. We illustrate this by applying it to a large online text-based therapy platform to characterize the development and perceived quality of the patient-therapist relationship.

Our analysis reveals that as their relationship develops, both patients and therapists redirect to a lesser extent, suggesting an overall smoother and more focused conversation flow. However, in relative terms, patients gain increasing control on the direction of the conversation over the therapists. Furthermore, we find that patients are more likely to eventually express dissatisfaction with the therapeutic relationship and terminate it when they are not able to redirect the conversation in the first few sessions. Cumulatively, these results underline the importance of considering patient agency in psychotherapy (Carey, 2010; Huber et al., 2021).

To summarize, in this work we:

- introduce a measure for quantifying the redirection effect of an utterance;
- apply this measure to a large online text-based therapy platform to characterize the development of therapeutic relationships;
- reveal a relation between the the extent to which a patient is able to redirect the conversation and their eventual dissatisfaction with therapeutic relationship.

While here we focus on the mental health domain, future work could apply this measure and our redirection framework to study the conversational process in other conversation-rich settings where complex long-term relationships are developed, such as education (e.g., advisor-advisee relationships). To encourage such work, we make our redirection framework publicly available.

2 Therapy Setting

Long-term text-based therapy. We develop our methodology in the context of Talkspace online therapy, a telehealth platform providing text based therapy from licensed therapists. Patients can

choose between different plans, which may include video therapies alongside messaging therapy. After a consultation in which they answer a few questions about their symptoms and state their preferences, patients are matched by the platform to a fitting therapist who is licensed in their state.

In this work, we use the (English-language) text-based conversation after personally identifiable information were removed by the platform.¹ Over the span of 5 years, the platform hosted over 18,000 licensed therapists which provided services to over 300,000 patients. In total, the dataset contains over 65 million messages. Therapies can be sustained over long periods of time, with over 26,000 therapies lasting over a year and 17,000 therapies having over 500 messages. This setting thus provides an unprecedented opportunity to study the long-term conversational process as the therapeutic relation develops.

At any point, patients can either cancel their therapy or switch their therapist. When doing so, they are asked to provide a reason, either by selecting from a drop down menu (e.g., "The treatment provided by my therapist was not helpful," "I met my goal / I feel better") or by entering free text. The provided reasons are not shared with the therapist, and the patients are informed of this confidentiality. These reasons serve as imperfect indicators on the quality of the therapeutic relation (for a broader discussion of difficulties in operationalizing the quality the therapeutic relation, see Section 8).

Sessions in text-based therapy. In studying the development of therapeutic relationships in this text-based setting, one challenge is accounting for the wide variety of duration, tempo, and volubility different therapies exhibit.

First, each therapist-patient pair develops a unique style and tempo of interaction, including frequency of interaction, how often they break up their message into multiple utterances, and response time. Second, text history between a therapist and a patient includes not only therapeutic conversation, but also short check-ins that deal with brief updates, scheduling, or survey completions. Last, an accurate picture of a text-based therapy is not a single long sequence of utterances, but multiple distinct synchronous conversations, or *sessions*, occurring over the duration of the therapy. A longitudinal analysis requires a methodology to capturing the

¹This research was approved by the IRB, and the use of the therapist and patient data is done with their consent.

progression of the therapeutic process in a way that allows for aggregation across therapies with vastly different interacting patterns. In particular, conventional units of progression in conversational analysis, such as time or number of utterances, are not well-suited for this complex instructional setting.

In this work, we center our analysis on *sessions* in order to capture the progression of the therapeutic relation while accounting for distinct dynamics. In text-based therapy, organization in sessions are not explicit, and thus we need to first automatically detect them from unstructured transcripts. We start by distinguishing high-activity periods interwoven with periods of little or no activity. We use the method from Kushner and Sharma (2020) to capture these bursts of activity in an unstructured transcript of patient-therapist interactions. We then define the session-split point as a moment when an utterance’s reply time has exceeded $N \times (\text{median reply time of that entire therapy})$, where N is a sufficiently large number (we settle on $N = 100$, see Appendix A.1 for details). We further ensure that each session represents an actual conversation between the patient and the therapist (as opposed to, for example, a short notification and acknowledgement on scheduling). After the split, if a given burst of activity does not have at least four turns taken—which requires at least two utterances from each speaker—or includes a video session, it is not considered a session and is discarded from our analysis. We further remove any automated messages related to video sessions, survey completion, or scheduling from the analysis.

To validate the quality of the session split, we sampled conversations that has at least 10 sessions and ran the Bayesian distinguishing-word analysis of Monroe et al. (2017) on the first and last utterance of a session from each speaker. The distinguishing words are consistent with an intuitive understanding: sessions starts with greetings ("hi," "morning," "hey," "how") and end with a farewell or expression of gratitude ("enjoy," "welcome," "thank," "thanks").

3 Redirection Measure

3.1 What redirection is (not)

We aim to quantify the redirection effect of a given utterance in a conversation: the extent to which it alters the immediate focus of the conversation. We start from the realization that for an utterance to

have a high redirection effect, it must (a) attempt to put the conversation on a different course (intention of redirection), and (b) receive a reply that is compliant with this redirection (realization of redirection). We will examine previous methods that capture these two conditions separately and explore how they guide us toward a design for a measure that jointly accounts for both of them.

Orientation. Orientation (Zhang and Danescu-Niculescu-Mizil, 2020) captures the degree in which an utterance *intends* to move the conversation away from what was already discussed. A "forwards-oriented" utterance (high orientation) is an utterance that intends to advance the conversation towards a specific target, and thus is *expected* to be followed by a reply centered around that target. A "backwards-oriented" utterance (low orientation) is an utterance that intends to address what was previously said, and thus is *expected* to follow a specific utterance. Thus, orientation characterizes an utterance’s intended objective (e.g., a forward-oriented utterance has a high intention of redirection), regardless of whether it is realized or not. In the first example of Figure 1, T_k is a high-orientation utterance since the therapist intends to redirect the conversation by asking the patient about specific source of stress at work. It so happens that in this example, the patient realizes this redirection in C_{k+1} by aligning themselves with the suggested direction, introducing their frustration with co-worker’s gossip. But even if they would have resisted the redirection attempt by reiterating, "I don’t know. I’m just stressed out all the time," (i.e., maintaining the focus on their general state of stress, rather than delving into specifics) the orientation of T_k would not be affected.

Similarity difference. To incorporate the actual reply, one may simply compute the similarity of the reply C_{k+1} with the original utterance T_k (following the notation in Figure 1) and compare it against a reference point to account for the ongoing direction of the conversation. A potential reference point is the extreme scenario when the therapist wishes to ensure no redirection will take place by simply repeating their previous utterance T_{k-1} . In other words, redirection of therapist’s utterance T_k could be formalized as the difference between the similarity of C_{k+1} and T_k and that of C_{k+1} and T_{k-1} .

Unlike orientation, this measure considers the actual reply of the utterance, but it hinges upon the assumption that semantic similarity can sufficiently

capture the redirection realization. Successful redirection, however, is not synonymous with similarity between the utterance and the reply. The two examples in Figure 1 illustrate this: while they intuitively show different levels of redirection, in both cases, the similarity difference is high.

Uptake. To account for a more nuanced take on redirection realization that goes beyond similarity, we could aim to measure the reply’s “uptake”: how much it answers, acknowledges, or builds upon the previous utterance. Demszky et al. (2021) formalizes this concept (in the context of student-teacher interactions) as the dependence of the reply on the utterance via point-wise Jensen-Shannon divergence, which uses next-sentence *prediction* to quantify the extent to which a given reply is a *probable* response.

While capturing redirection realization better than the similarity measure, uptake does not account for the redirection intent. For instance, both examples in Figure 1 show high level of uptake with the patient replies being consistent with the respective therapist utterances, but different levels of redirection because only in the first example, the intent of the therapist was to move the conversation away from what was previously discussed.

Our approach: Redirection. We combine the insights of these existing measures to develop a new redirection measure that captures both the intention and realization components. In particular, our analysis of the previous measures points toward the need to consider the actual reply the utterance receives, to use a predictive component to capture the extent to which the reply naturally follows the utterance, and to use a point of reference to capture a change in direction.

More concretely, to quantify the redirecting effect of an utterance, we first consider the likelihood of the patient’s reply given the previous context as a measure of the redirection realization:

$$\mathcal{P}_k(C_{k+1}) \triangleq P(C_{k+1}|C_k, T_k).$$

We condition on the most recent utterances from both speakers, C_k and T_k , to capture both of their prior conversational context.

And as a point of reference, we consider the likelihood of the reply given the therapist repeated their previous utterance—an extreme hypothetical scenario in which the therapist wishes to ensure no redirection will take place by simply repeating their previous utterance:²

²One naive alternative might be to calculate the $\mathcal{Q}_k(C_{k+1})$

$$\mathcal{Q}_k(C_{k+1}) \triangleq P(C_{k+1}|C_k, T_{k-1}).$$

We then formally define the *redirection* of T_k as the log-odds ratio of these two probabilities:

$$R(T_k) \triangleq \log \left(\frac{\mathcal{P}_k(C_{k+1})}{1-\mathcal{P}_k(C_{k+1})} / \frac{\mathcal{Q}_k(C_{k+1})}{1-\mathcal{Q}_k(C_{k+1})} \right) \quad (1)$$

When redirection is high, $\mathcal{P}_k(C_{k+1}) \gg \mathcal{Q}_k(C_{k+1})$, T_k alters the direction of the conversation, in the sense that the patient’s reply C_{k+1} is likely as a reply to this utterance, but not as a continuation of the previous direction of the conversation. Conversely, in cases of low redirection, the patient’s response is less affected by the therapist’s utterance, as it would be likely even in its absence.

We compute the redirection of a patient’s message in a corresponding way.

Operationalization. To compute the probabilities for our redirection measure, we fine-tuned Gemma-2B model (Team et al., 2024) with 4-bit QLoRA (Detmers et al., 2023) using a dataset of 8,000 therapy conversations.³

We obtain a model adapted to therapy conversations that can more accurately capture the interactions between patients and therapists. We use this fine-tuned model to compute the probability terms.

The operationalization for related measures is discussed above are included in the Appendix B.1.

3.2 Session-level aggregation

To apply and compare our utterance-level measures across therapy sessions, we use two aggregation methods. The first method averages the utterance-level measures per session for each speaker. A speaker’s average redirection in a session is computed as the mean redirection of their n utterances in that session, which we denote with T_{avg} and C_{avg} for the therapist and patient respectively.

The second method examines the balance of redirection between speakers to determine who is redirecting the conversation more in a given session. Analyzing the balance of redirection can reveal how the control of the discussion is shared between the patient and therapist. We consider relative measure

without conditioning on the previous therapist utterance T_{k-1} , to simulate the “absence” of any therapist utterance. This alternative, however, creates an unnatural situation with the patient replying to themselves (which would artificially have a low probability). Furthermore, such an alternative would not be directly comparable with $\mathcal{P}_k(C_{k+1})$, which is conditioned on the previous utterances of both speakers. Therefore, we employ a reference point with equivalent conditioning.

³The models were trained on internal GPU servers in consideration of the sensitive nature of the data. Training details are included in the Appendix B.1.

as function of the average redirection measures of both the therapist T_{avg} and patient C_{avg} in a session. Equation 2 formalizes relative redirection from the therapist’s perspective:

$$T_{rel} = \frac{\exp(T_{avg})}{\exp(T_{avg}) + \exp(C_{avg})}. \quad (2)$$

The relative value for the patient C_{rel} is computed in a corresponding way. Note that mathematically:

$$T_{rel} + C_{rel} = 1. \quad (3)$$

4 Redirection in Online Therapy

We employ our formalism to study redirection in online text-based therapies. We first validate that our redirection measure corresponds to human intuition in the therapy setting. We then explore how redirection relates to the development of the therapeutic relation. Finally, we explore the relation between redirection and the patient’s perception of the quality of the therapy, in addition to discussing the related measures discussed in Section 3 to gain insight into which interactional dynamics might explain the observed trends.

Human validation. To check how aligned our measure is with our intuitive understanding of redirection in this particular online therapy setting, we designed a redirection identification experiment. A participant is shown a pair of two short interactions, each consisting of 4 utterances (like the ones in Figure 1). Their task is to pick which one of them has a higher redirection effect in the second to last utterance (e.g., T_k in Figure 1).

Each pair for this experiment is constructed by first picking a random therapy, selecting the utterance with the highest and lowest redirection effect according our measure, and considering the interaction surrounding those utterances. We select 10 such pairs using our redirection measure, and for comparison we select 10 more pairs for each of the other measures discussed in Section 3. If a measure properly captures the redirection effect, we expect it to match the human rankings.

To respect the privacy of the data and adhere to IRB protocol, the task was performed by one of the authors who was authorized to read the therapy text. To avoid author bias, we administer the task in a blind fashion, with the participant not knowing which pairs were selected using which measure.

The participant ranking perfectly matched that of our redirection measure, compared to 8 out of 10

matches for the similarity difference, 5 out of 10 for orientation and 4 out of 10 for uptake. While limited in scale due to the privacy restrictions on the data, these results suggest that our measure does indeed capture an intuitive notion of redirection which the other measures do not. The experiment also inspired a qualitative analysis comparing examples in which the metrics disagree (Section 5).

Shuffling check. The phenomenon that we seek to capture is inherently reflecting conversational dynamics, and thus it should be tied to the specific order in which the utterances appear in the conversation. To check if what we are indeed capturing is a conversational phenomenon, we artificially break the order of each session by shuffling its utterances. We would expect no redirection to occur in this original setting.

Indeed, while the average redirection across all (non-shuffled) sessions is 6.40, the average redirection in the shuffled sessions is near zero (-0.025 ; $p < 0.0001$ for the difference according to a Wilcoxon signed-rank test).

In our main analysis below, we will use the shuffle check to ensure that the trends we observe are actually related to conversational processes and not to spurious non-conversational effects (e.g., a change in the length of the session or in the length of the utterance).

Evolution of the therapeutic relation. We are now ready to proceed to our main analysis: examining the relation between redirection and the development and quality of the therapeutic relation. We start with examining the change in redirection as the therapist-patient relation develops, focusing on sustained therapies having at least 10 sessions (3,764 in total).

Figure 2 shows the average redirection measure T_{avg} and C_{avg} in each session for the first 5 and last 5 sessions. For both the patient and therapist, there is a downward trend in local redirection as therapy progresses ($p < 0.0001$ according to a Wilcoxon signed-rank test comparing the first and last 5 sessions). Thus, as the therapy progresses, redirection in conversation occurs less from both speakers, perhaps suggesting a smoother flow of the conversation with more stable conversational goals (see Section 5 for a qualitative analysis).

We now switch from considering each speaker’s redirection separately to examining how the *balance* between the two evolves over time. Figure 3 shows the patient’s relative redirection C_{rel} across the first 5 vs. last 5 sessions. As the relation pro-

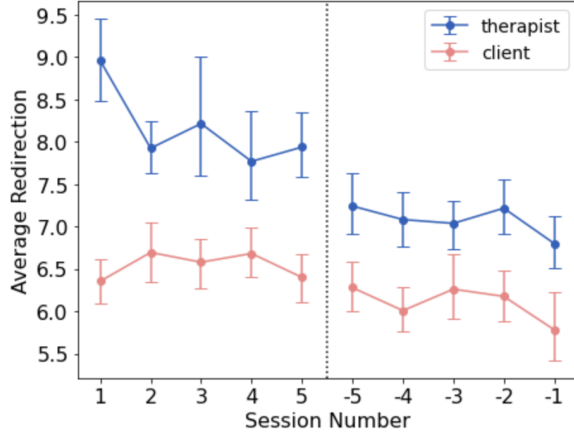


Figure 2: Average redirection across the first 5 and last 5 sessions. Throughout, error bars indicate 95% confidence intervals estimated through bootstrap sampling.

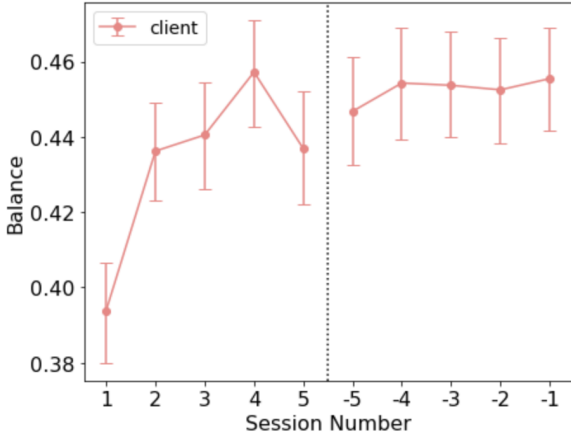


Figure 3: Patient relative redirection across the first 5 and last 5 sessions.

gresses, the patient share of the redirects increases over time ($p < 0.0001$), suggesting that the patients are gaining more control over the flow of the conversation from the therapists.

We want to ensure that these trends capture redirection as a measure of conversation dynamics and not a spurious byproduct of other session-level trends—for example, the number of utterances per session and the number of tokens per utterances, consistently decrease as therapy progresses. We re-analyze the trends after shuffling the utterances within a session. In doing so, all session-level trends are maintained with only the immediate interaction of the two speakers within a session disrupted. After the shuffle, all observed trends in the original data disappear (no statistical difference), suggesting that they are indeed tied to the conversation dynamics.

Unsuccessful relationships. We now investigate

how redirection is related to the quality of the therapeutic relationship. For this purpose, we consider a therapeutic relation to be “unsuccessful” if the patient eventually abandons it via a request to switch the therapist or to cancel the subscription *and* provides a reason that explicitly cites dissatisfaction with the therapist or with the relation (see Appendix A.3 for a list of such reasons). After filtering out therapies that had less than 3 sessions (to ensure that the patient actually “gave it a try”), we are left with 817 such unsuccessful relations. For a meaningful comparison, we consider a control group with the same number of therapies where the patient did not request to cancel or switch the therapist, and that also has at least 3 sessions. To discard effects due to the duration of the therapy and focus on signals that could be perceived early in the therapeutic process, we only consider the first 3 sessions in this analysis. We find that in unsuccessful relationships, patients redirect the conversations less than in the control group ($p = 0.018$), while we do not find any difference for therapist ($p = 0.5$). We note these differences disappear after shuffling the order of the utterances in the session ($p = 0.17$), thus they are reflecting differences tied to conversation dynamics.

Insights from other measures. We also explore how our redirection measure compares itself from the other related measures discussed in Section 3. The results are summarized in Table 1 in the appendix C. It is worth noting that none of the related measures show a significant difference between the “unsuccessful” therapies and the control group, and only the similarity difference measure exhibits temporal trends that pass the shuffling test.

We highlight here the results for orientation, since by capturing the redirection intent separately, they provide further context for interpreting the results discussed above. Orientation shows a significant downward trend for therapist, thus observed decrease in redirection can be at least partially attributed to the decrease in their attempts to do so (rather than on the patient’s unwillingness to realize those attempts). Furthermore, there is no difference in patient orientation when comparing “unsuccessful” relations with the control group ($p = 0.5$), suggesting that reduced patient redirection is not there for a lack of redirection bids, but perhaps because of the therapist not accepting those bids. Further work could strengthen this suggestive evidence by more directly separating the intention and realization component of redirection.

5 Qualitative Analysis

We conduct a qualitative analysis of high- and low-redirection examples to explore therapy strategies that are tied to this phenomenon and to further interpret the observed trends. Additionally, we analyze examples where the redirection measure deviates from the related measures to check our intuition about differences in what they capture. The discussion follows examples included in Table 2 in the Appendix.

Redirection strategies. In our setting, highly redirecting utterances of a therapist typically involve exploration and surveying. Therapists tend to use open-ended questions to assess their patients' given situation and viewpoints (Example 1: "*Where does this [issue] come from*"). They may also suggest different perspectives or provide guidance on addressing these issues (Example 2: "*Try doing ...*"). Conversely, less redirecting utterances tend to echo or validate the patient's statements or empathize with their struggles. They may also reformulate what the patient has said and keep the focus on the immediate problem at hand (Example 4: "*Sounds awful. Is it...*").

For patients, highly redirecting utterances tend to introduce personal experiences or feelings that shape the immediate course of the conversation. They may initiate subjects they wish to explore or verbalize the specific obstacles they are facing (Example 6: "*Challenges at work*"). Low patient redirecting utterances, on the other hand, extend the current subject of discussion.

These observations are consistent with the decreasing trend in redirection for both speakers. Initially, patients and therapists both focus on introducing the context of the therapy and setting therapeutic goals. Therapists tend to actively discuss the objectives of the therapy and suggest behavioral strategies and cognitive techniques they will pursue throughout the therapy. Patients, likewise, will share their background or expectations as they start out their therapeutic relationship. Latter sessions, conversely, usually consist of more concentrated discussion on the identified issues.

Comparison with other metrics. We also examined cases where the redirection measure is at odds with the related measures. As expected, orientation can be high even when redirection is resisted. For instance, in Example 3, the therapist attempts to steer the conversation towards discussing who to talk to, but the patient disregards their sugges-

tions to communicate and continues explaining their thoughts and plans, thus causing redirection to be low.

Importantly, unlike the similarity difference approach, our metric captures utterances with redirection effects even in cases when the reply is not semantically similar. In Example 2, the therapist redirects the focus of the conversation by suggesting a potential solution to the patient's problem, which the patient acknowledges in their reply. While similarity difference is low since acknowledging the suggestion is not semantically similar, the patient does reflect on the therapist's suggestion, which deviates from their prior discussion focused on the problem.

Our measure also differs from uptake in that it uses a reference point—the previous conversational context. Example 7 illustrates a segment where the the patient reflects on having someone to vent to. The therapist addresses the patient's utterance by continuing to encourage the patient to vent, exhibiting high uptake. However, the focus of the conversation does not change, indicating low redirection.

6 Further Related Work

Our work relates to prior research on conversational dynamics, analyzing the development and quality of therapeutic relations, and exploring language in mental health platforms.

Development and quality of therapeutic relation. The development of the patient-therapist relation and its impact on therapy outcomes have also been extensively studied (Gelso and Carter, 1985; Norcross, 2010). Previous work characterized therapeutic alliance (Goldberg et al., 2020) and ruptures (Tsakalidis et al., 2021) through emotional engagement (Christian et al., 2021), sentiment (Syzdek, 2020), linguistic coordination (Nasir et al., 2019), and synchrony (Doré and Morris, 2018). Our work characterizes an additional dimension of the patient-therapist based on the joint act of redirection.

Conversational dynamics. Prior computational work examined different aspects of conversation flow (Zhang et al., 2016), including work on topic segmentation (Eisenstein and Barzilay, 2008; Nguyen et al., 2014; Purver, 2011; Glavas and Somasundaran, 2020; Jiang et al., 2023) and topic shift (Xie et al., 2021). While these conversation-level concepts are related, our probabilistic mea-

sure seeks to quantify the immediate effect of a single utterance, an effect that goes beyond semantic similarity or topic coherence. Furthermore, we develop our redirection framework in the distinctive domain of online mental health therapy.

Online mental health platforms. Prior literature extensively explores linguistic behaviors and conversational choices users make in online mental-health related platforms, whether in crisis counseling platforms (Althoff et al., 2016; Zhang and Danescu-Niculescu-Mizil, 2020), peer-support platforms (Pruksachatkun et al., 2019; Yang et al., 2017), or therapy platforms (Malgareoli et al., 2023b). Accompanied with growing attention to online mental health resources and use of technology in psychotherapy (Anthony, 2003; Barak et al., 2008), many studies highlighted the benefits of online social support (De Choudhury and Kiciman, 2017; Newman et al., 2011). We specifically focus on *long-term* sustained relationships in text-based therapy platforms, offering a novel perspective on development of patient-therapy relation through longitudinal analyses.

7 Discussion

Redirection is a joint act often performed by therapists and patients. Its realization requires both speakers to understand the initial direction of the conversation, one’s attempt to redirect it, and the other’s compliance to this attempt. As such, its study has the potential to characterize the patient-therapist dyad, in particular with respect to their ability to negotiate the focus of the conversation.

In this work, we introduce a computational method for quantifying the redirection effect of an utterance and apply it to a mental health therapy domain. We find that both the patient and the therapist redirect less as therapy progresses. Our qualitative analysis suggests that as the therapeutic relations mature, the patients and therapists are able to locate the specific focus of their therapy, after initial exploration and contextualization prompted by both speakers. Moreover, we reveal that a low levels of redirection initiated by the patient can signal failure in maintaining a therapeutic alliance, indicated by the patient eventually expressing dissatisfaction and abandoning further therapy with the therapist.

Relation with psychotherapy literature. Several decades of research in psychotherapy suggest that the “treatment model” approach in which an ex-

pert clinician provides curative treatment that the patient passively receives is not well supported by outcomes or engagement data found in psychotherapy (Bohart, 2000; Duncan and Miller, 2000). Instead, shifting focus from which aspects of treatment are delivered to how dyads collaboratively negotiate therapeutic conversations has provided evidence of improved access, efficiency, and effectiveness of mental health services for “patient-led” approaches (Carey, 2010; Carey et al., 2013; Huber et al., 2021). Our results here offer a computational perspective into one dimension of patient agency in therapeutic relationships.

Future work. A computational approach to redirection enables us to observe the evolution of therapeutic relations and could assist therapists in fostering them. A complementary line of work can include a more in-depth exploration on how to foster healthy levels of redirection in therapy and the causal relationship between redirection and the quality of therapeutic relationships. For instance, labeling messages with therapeutic strategies and techniques can provide insight into the effectiveness of each strategy and determine which ones are more applicable for specific contexts (e.g., when patients resist therapists redirection attempts).

In their current observational form, these results suggest that early conversational patterns can signal the eventual dissatisfaction of the patient. Future work could examine the predictive power of these ties and test the extent to which they might be explained by other (unalterable) factors, such as the characteristics of the patient or of their condition.

Our methodology can extend beyond the mental health domain and be applied to examine other conversation rich domains, such as education, online discussions, or interviews. These settings are where the discussions are carried in relatively unstructured ways with only a few general agendas set. Explorations of redirection in these contexts may present a unique perspective into how speakers are able to redirect and control the flow of the discussions.

Finally, understanding how humans redirect the flow of conversations is important for supporting more naturalistic human-AI conversations, where the AI can pick up on humans’ redirection attempts and initiate their own.

8 Limitations

In the mental health context, how to define successful therapeutic relations is always open. Our work uses patient-provided reasons for canceling the therapy or switching the therapist that explicitly mention dissatisfaction with the therapist as an imperfect indicator for failure of the therapeutic relationship. A reliable positive signal, on the other hand, is difficult to define, especially in a therapy domain where patients can have hidden agendas, present misleading information, or choose to remain in treatment for a number of reasons (Newman and Strauss, 2003). We thus do not consider the control group to necessarily contain “successful” relations, and thus the interpretation of the results of the comparison should be interpreted accordingly.

Furthermore, these results are purely observational, and future work would be needed to establish if intervening to change the amount of realized redirection (e.g., through training of the therapists) would have an effect on the quality of the therapeutic relationship.

The notion of session we employed captures the unique structure of therapy that arises as patients and therapists maintain a long term relation spanning multiple conversations, rather than just a single encounter. However, sessions’ characteristics change as time progresses. We observed that the length of the session and the token count of an utterance in a session both get shorter. Therefore, as a unit of progression, it has the shortcoming of having variability over time. While our shuffling experiments suggest that our observations are not a mere result of this variability and are tied to the actual dynamics of the conversation, further work is needed to fully account for this variability.

Our work focuses on text-based conversations, and while these are a substantial component of online therapy, they do not capture the occasional video conversations that patients and therapists might have. Future work could explore analyzing the transcripts of these video conversations, and the role they have in the development of the relationships (although additional de-identification challenges arise when working with video data).

As the redirection measures actualized change in subject of the conversation, it requires the reply of the utterance to compute. This constraint can be restrictive in practice, as an automated system would be unable to analyze an ongoing conversation with-

out a reply it hasn’t received yet. Exploring how we can predict whether the upcoming reply will cooperate with one’s attempt to redirect remains an interesting direction for future work.

Ethical Concerns. As the work involves highly sensitive data in the form of patient’s chat therapy history and surveys, all personally identifiable information was removed. Access to data was strictly limited to the authors of the paper, and data was processed on restricted servers and will be removed upon publication. All large language models trained on the data were strictly internal and discarded after the analysis.

This work is done in close collaboration with the therapy platform. All participants consent to the use of their data in aggregate and de-identified format for research purposes as part of the Terms of Service they review during onboarding in the case of the patients, or they review it during the hiring process in the case of the therapists. All individuals may opt out of the use of their data for research purposes at any time without penalty.

Given the sensitivity of the domain and the limitations discussed above, extensive work is still required before insights from our results could be used in patient-facing systems.

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A Additional Data Details

A.1 Session Split

Adapting the methodology in [Kushner and Sharma \(2020\)](#), we chose a value of N for the session split based on the number of sessions it produces for 10,000 random conversations. The median, mean, and standard deviation all plateau starting from $N = 50$ and level off around $N = 100$. Setting N at 100 will provide the reliability of the measure in relation to the selection of its value.

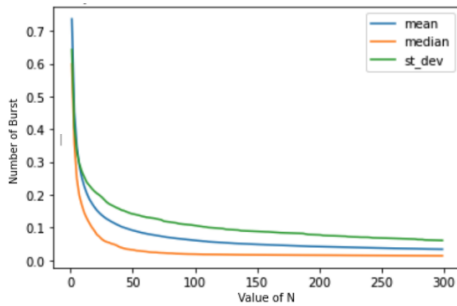


Figure 4: Number of sessions per N from 10,000 random conversations.

A.2 Switch / Cancel Surveys

During therapy, patients may choose to switch therapists or cancel their plan by filling out surveys. In the survey, they are asked to provide a reason for the switch/cancel, which they either select from a fixed list, or enter their own in free text. The provided switch reasons include:

- s1 I could not find a time to meet with my provider.
- s2 I couldn't form a strong connection with my provider.
- s3 I don't feel like my provider was responsive enough.
- s4 I just want to try someone new.
- s5 I want to select a provider with a different gender.
- s6 I was unsatisfied with the quality of care.
- s7 Disabled.
- s8 Dissatisfied with app.
- s9 Dissatisfied with provider.
- s10 Expensive.

The provided cancel reasons include:

- c1 I met my goal / I feel better.
- c2 The cost doesn't fit my budget.
- c3 The treatment provided by my therapist was not helpful.
- c4 I had technical issues.
- c5 My therapist was not responsive to my messages.

A.3 Defining Unsuccessful Relations

The dataset of "unsuccessful" therapies with the patient abandoning the relationship includes therapies where the patient requested either a switch with reasons s2, s3, s4, s6 or a cancellation with reason c3.

B Operationalization

B.1 Implementation Details

For our redirection model, we fine-tuned Gemma-2B ([Team et al., 2024](#)) with 4-bit QLoRA ([Dettmers et al., 2023](#)) using the huggingface library. We use a 90/10 split for training and validation, and trained for 2 epochs, with LoRA rank = 16 and dropout = 0.05, context length 4096, batch size 2, learning rate $2e-4$, AdamW optimizer ([Loshchilov and Hutter, 2017](#)), achieving a validation perplexity of 10.97. The total train time is approximately 43 hours on 2 NVIDIA RTX A6000 GPUs. For the model experiments, we conducted a hyperparameter search over learning rates [$2e-5$, $2e-4$] and LoRA rank = [8, 16, 64], and used fixed values for the rest of the parameters.

We used ConvoKit Python package ([Chang et al., 2020](#)) to calculate orientation to employ the same methodology outlined in [Zhang and Danescu-Niculescu-Mizil \(2020\)](#). We trained two separate models for therapist and patient orientation, using dependency-parse arcs representations for both speakers and 12 SVD dimensions.

To calculate uptake, we fine-tuned a pre-trained BERT-base model for next utterance classification. Two separate models were trained: one for predicting patient's utterance after a therapist and the other for vice-versa. Our training setup follows the original paper ([Demszky et al., 2021](#)). We fine-tune our model for 1 epoch with a batch size of 16, max length of 512 tokens for patient's and therapist's utterance each. The learning rate is set

at 6.24e-5, with linear decay and AdamW optimizer.(Loshchilov and Hutter, 2017) The total train time is approximately 12 hours for each model on 2 NVIDIA RTX A6000 GPUs. We used the two models with the original source code from (Demszky et al., 2021) to calculate uptake.

For similarity difference, we used a pre-trained sentence BERT model 'multi-qa-MiniLM-L6-cos-v1' to map utterances into a 384 dimensional dense vector space. We calculated the similarity between two utterances using cosine similarity of their embeddings.

B.2 Used Artifacts

We list the following artifacts and their licenses that are used in the work.

- ConvoKit 2.5.3:
<https://convokit.cornell.edu/>, MIT License
- PyTorch 2.2.1:
<https://pytorch.org>, BSD-3 License
- Sentence Transformers 3.0.0:
<https://github.com/UKPLab/sentence-transformers>, Apache License 2.0
- Transformers 4.38.2:
<https://github.com/huggingface/transformers>, Apache License 2.0
- Conversational Uptake Source Code:
<https://github.com/ddemszky/conversational-uptake>, MIT License

C Related Measures Results

Table 1 summarizes the results from redirection and other measures – orientation, uptake, and similarity difference.

D Qualitative Examples

Table 2 gives 8 examples of therapeutic interactions where the redirection measure differs from other related measures.

Analysis	Redirection	Orientation	Similarity Difference	Uptake
Start/end difference in average value	↓ / ↓	- / ↓	↓ / ↓	↑ / ↑
Start/end difference in balance	↑ / ↓	↑ / ↓	- / -	- / -
Maintains trend af- ter shuffling	No	Yes	No	Yes
Distinguish unsuc- cessful relations	Yes	No	No	No

Table 1: Summary of results for four different measures related to redirection. For cells with two results divided by /, each one corresponds to patient / therapist respectively. ↑ indicates a higher level in the end than the start; ↓ indicates vice-versa. (Cells with - indicate no significance.)

#	Example	R	O	SD	U
1	T: It seems like this anxiety doesn't arise when you're in a relationship with someone. P: Now that I remember no. I feel more secure. T: Where does the strong reaction to being alone originate from? P: Honestly, I think it has to do with my relationship with my family.	High	Low	Low	Mid
2	T: It seems like the issue is your lack of control over being alone. P: Yes, that's interesting; I think that's a part of it. T: Try sitting with your feelings and exploring them without blocking them, and see what insights come up for you. P: Okay, I'll meditate on that. Thanks.	High	Mid	Low	Mid
3	T: I believe using conflict resolution methods might help you better communicate with your husband about this issue, rather than withdrawing and shutting down. P: He keeps saying he doesn't want to get involved with her problems. T: Then, are you planning to talk to your daughter first, or are you waiting to speak with her counselor? P: I'll just ground her and take things away.	Low	High	Low	Mid
4	T: When you are ready to process the trauma, we can go through it. P: Now, I'm feeling better mentally, but physically my heart is thumping, and I began having panic attacks at work. T: That sounds awful. Is it a thumping sensation in your chest or feeling in your ears? P: I wore a heart monitor in my chest. I feel a thump each time before my heart skips beats.	Low	Low	High	High
5	P: I dislike how I get easily attached to a man I like. I had thoughts of being in a relationship just because we were spending time with each other. T: Some people are attachers in relationships, while some people it takes a while for the heart to thaw out. P: I feel like it may be because of not having my dad present. T: I think that is a good insight on your part. Your upbringing and personality formation definitely affects attachment styles.	High	Low	Low	Low
6	P: How was your day? T: I'm also waiting for warm weather; I think today's supposed to be pretty nice! How are things going at home for you? P: It's going okay. I'm nervous about having to hit numbers at work every day though. That is going to be a challenge. T: I understand productivity requirements. It sounds like you have an amazing work ethic and your trainer is already impressed with you though.	High	High	Mid	Low
7	P: I'm going to take a walk tonight if weather permits. T: The walk sounds like a great idea. It is good to let it all out, so don't be afraid to vent as much as you need to. P: Thanks. Not used to having someone to vent to. T: Hope the rest of the day went better and the weather allows you to take a walk. Feel free to vent as things come up.	Low	Low	Low	High
8	P: When I have trouble concentrating, I tend to eat. T: Does that help you concentrate better? P: I don't know. T: So what's behind eating when you are having trouble concentrating?	Low	Low	Low	Mid

Table 2: Qualitative examples comparing Redirection (R) with other measures: Orientation (O), Similarity Difference (SD), and Uptake (U). The measure is applied to the colored utterance (Therapist for the first 4 examples and Patient for the remaining 4). Low indicates the bottom 25th percentile of the measure values for the respective speaker type, High indicates the top 75th percentile, and Mid is 25-75th percentile. Examples are paraphrased to preserve privacy.